

ECO 202: Statistics and Data Analysis for Economics – Fall 2026

Princeton University

Last update: August 30, 2026

1 Introduction

This undergraduate class develops the probability and statistical foundations needed for empirical work in economics and for later study of econometrics. Topics include descriptive statistics, probability, sampling, estimation, confidence intervals, hypothesis testing, causal inference, and an introduction to linear regression. The course emphasizes statistical reasoning, interpretation, reproducible empirical work, and the productive, ethical, and critical use of AI at the interface of economics and data science.

2 Useful Information

Instructor: Matias D. Cattaneo (cattaneo@princeton.edu)

Lectures: Mon and Wed, 1:20 pm – 2:40 pm, Robertson 100

Class Website: <https://canvas.princeton.edu/>

GitHub: <https://github.com/mdcattaneo/ECO-202>

TAs: Ryan Broll (rb0555@princeton.edu)

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3 Course Format and Use of AI

ECO 202 is an instructor-led, AI-integrated course. Each instructional class meeting is organized around a Markdown guide available in the course GitHub repository. These guides provide the stable statistical spine for preparation, class, and review; they do not replace lecture attendance and discussion. During class, I will use each guide to navigate the material, work through important statistical reasoning, run selected data demonstrations in R, and at selected points interact with an AI system in view of the class. We will use generated explanations, calculations, code, graphs, and interpretations as objects to inspect rather than as automatically correct answers. I remain responsible for the course narrative and pace, while students and I remain responsible for the statistical judgments we make.

The reference classroom workflow places the Markdown class guide and an AI assistant side by side in Visual Studio Code, with R used for data analysis. This is an example rather than a requirement. Students may choose the AI system, model, interface, editor, and statistical software that best fit their access, preferences, and goals. R will be used in lecture, but students may work with R, Python, Stata, or another suitable environment.

Students are encouraged to use AI while preparing for class, studying the Markdown guides, and working on ungraded practice. Productive uses include retrieving prerequisites, requesting a staged hint, comparing explanations or solution strategies, generating a nearby practice question, proposing or critiquing code, diagnosing a graph or empirical claim, and reconstructing a solution after an attempt. Students should make a genuine attempt before requesting a complete solution and should preserve regular opportunities to work without assistance. The repository's [course AI guide](#) and folder-level guides provide reusable prompts and activity-specific protocols. These guides support learning but do not override this syllabus or an explicit instruction given in class.

AI output is neither an empirical fact nor a scholarly or statistical authority. Students are responsible for checking its assumptions, calculations, code, graphs, citations, and conclusions against the course references, executed analysis, their own reasoning, and other reliable cited sources. Generated material must not be represented as the student's independent reasoning or as an instructor-approved solution. Instructions for each submitted or evaluated activity will state whether AI is permitted and what disclosure is required.

The four in-class exams are closed-book and fully unaided: AI systems, electronic devices, notes, books, and outside assistance may not be used except for an accommodation authorized by the University. The exams assess the same standard of independent statistical mastery expected before generative AI. Students must be able to formulate statistical questions, perform essential calculations, state assumptions, select and justify methods, interpret evidence, distinguish association from causation, and identify limitations on their own. The exams do not assess memorized software syntax, installation steps, or features of a particular AI product.

The individual empirical project will permit and actively support AI use under project-specific requirements for documentation, reproducibility, verification, attribution, privacy, interpretation, and individual responsibility. AI may assist with brainstorming, research design, data work, coding, visualization, interpretation, and editing, but each student remains responsible for running and auditing the analysis and for every submitted choice, claim, computation, citation, and interpretation. A key goal of this class is to help students develop a productive and ethical AI-supported workflow.

No required learning activity depends on a paid AI service or on creating an account with a third party. Every required activity has a non-AI route through the class guides, lectures, textbook and other references, and instructor or TA support. Students should not enter private student work, grades, accommodations, personal information, confidential or restricted data, or course-restricted copyrighted material into an external AI system. Princeton University policies and course-specific instructions concerning academic integrity, privacy, and intellectual property apply throughout.

4 Grading

The grading will be based on four in-class, closed-book exams and an individual empirical project. Each component will contribute 20% towards the final numerical grade. AI assistance is not permitted during the four exams. The exams assess independent command of the common statistical foundation; the project assesses the ability to carry out and communicate a responsible empirical analysis using modern tools.

A tentative numerical-to-letter-grade mapping is available by default in Canvas and follows Princeton University's recommended grading practices and policies. I reserve the right to alter this mapping without notice.

5 Class Schedule and Important Dates

Date	Topic
Wed 2-Sep	Course Introduction, AI-Assisted Learning, and Modern Data Workflows.
Mon 7-Sep	<i>Labor Day holiday—no class.</i>
Wed 9-Sep	Descriptive Statistics and Data Visualization.
Mon 14-Sep	Density Curves, Normal Distributions, and Standardization.
Wed 16-Sep	Scatterplots, Correlation, and Descriptive Regression.
Mon 21-Sep	Association, Causation, and Confounding.
Wed 23-Sep	In-Class Exam 1.
Mon 28-Sep	Populations, Samples, Surveys, and Selection Bias.
Wed 30-Sep	Randomized Experiments, Observational Studies, and Causal Effects.
Mon 5-Oct	Probability Rules and Probability Models.
Wed 7-Oct	Conditional Probability, Independence, and Bayes' Rule.
Mon 12-Oct	In-Class Exam 2.
Wed 14-Oct	<i>University examination week—no class.</i>
Mon 19-Oct	<i>Fall recess—no class.</i>
Wed 21-Oct	<i>Fall recess—no class.</i>
Mon 26-Oct	Random Variables, Expectations, Variance, and Covariance.
Wed 28-Oct	Sampling Distributions for Counts, Proportions, and Means.
Mon 2-Nov	Laws of Large Numbers and the Central Limit Theorem.
Wed 4-Nov	Point Estimation, Bias, Variance, and Standard Errors.
Mon 9-Nov	In-Class Exam 3.
Wed 11-Nov	Hypothesis Tests, p -Values, Statistical Significance, Errors, and Power.
Mon 16-Nov	Confidence Intervals and Their Connection to Hypothesis Tests.
Wed 18-Nov	Inference for and Comparison of Means and Proportions.
Mon 23-Nov	<i>Thanksgiving recess week—no class.</i>
Wed 25-Nov	<i>Thanksgiving recess week—no class.</i>
Mon 30-Nov	Conditional Distributions, Expectations, and Simple Regression.
Wed 2-Dec	Multiple Regression, Causal Interpretation, and Project Workshop.
Mon 7-Dec	In-Class Exam 4.
Sun 20-Dec	Final Assessment: Individual Empirical Project due at 7:00 pm.

6 Individual Empirical Project

Each student will complete an individual empirical project in lieu of the final examination. The student will formulate an empirical question, analyze data using appropriate course methods, and communicate the findings and limitations clearly. Each student is responsible for every submitted choice, claim, computation, citation, and interpretation, and for making the analysis reproducible.

AI use is permitted and encouraged as part of the project workflow. It may assist with brainstorming, research design, data work, coding, visualization, interpretation, and editing, but its output must be audited rather than accepted automatically. Students will not be graded on their choice of AI vendor or statistical software, or on how much programming syntax they write without assistance. Evaluation will focus on the quality of the empirical question, data and methods, statistical reasoning, verification, reproducibility, communication, and honest documentation of the process.

The project is due at 7:00 pm on December 20, 2026. Detailed guidance will specify the question and data requirements, milestones, deliverables, AI-use record and disclosure, reproducibility standard, attribution and privacy rules, grading criteria, and submission procedure.

7 References

Official textbook listed for the course:

- Moore, D. S., G. P. McCabe, and B. A. Craig (2021): Introduction to the Practice of Statistics, 10th edition, Macmillan Learning.

This book is the official reference, but the course will not follow it mechanically or treat it as the exclusive source of explanations and practice. The following references offer alternative explanations, examples, applications, and levels of depth. Students may consult whichever resources best support their learning; none is an additional required textbook.

1. Diez, D. M., M. Çetinkaya-Rundel, and C. D. Barr (2019): OpenIntro Statistics, 4th edition, OpenIntro.
2. Çetinkaya-Rundel, M. and J. Hardin (2024): Introduction to Modern Statistics, 2nd edition, OpenIntro.
3. Spiegelhalter, D. (2019): The Art of Statistics: Learning from Data, Basic Books.
4. Wooldridge, J. M. (2020): Introductory Econometrics: A Modern Approach, 7th edition, Cengage.
5. Stock, J. H. and M. W. Watson (2019): Introduction to Econometrics, 4th edition, Pearson.
6. Huntington-Klein, N. (2025): The Effect: An Introduction to Research Design and Causality, 2nd edition, Chapman & Hall/CRC.
7. Wickham, H., M. Çetinkaya-Rundel, and G. Grolemund (2023): R for Data Science, 2nd edition, O'Reilly Media.